



Rental Application Processor

Automating the Rental Application Workflow Using RPA and AI-Driven Agents

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The Problem: Dublin's Rental Market

€2.2–2.6K

Monthly Rent

Two-bed inner Dublin (RTB Q4 2024)

80–120

Applications

Per listing in under 48 hours

6–8hrs

Agent Workload

Manual review per property

20min

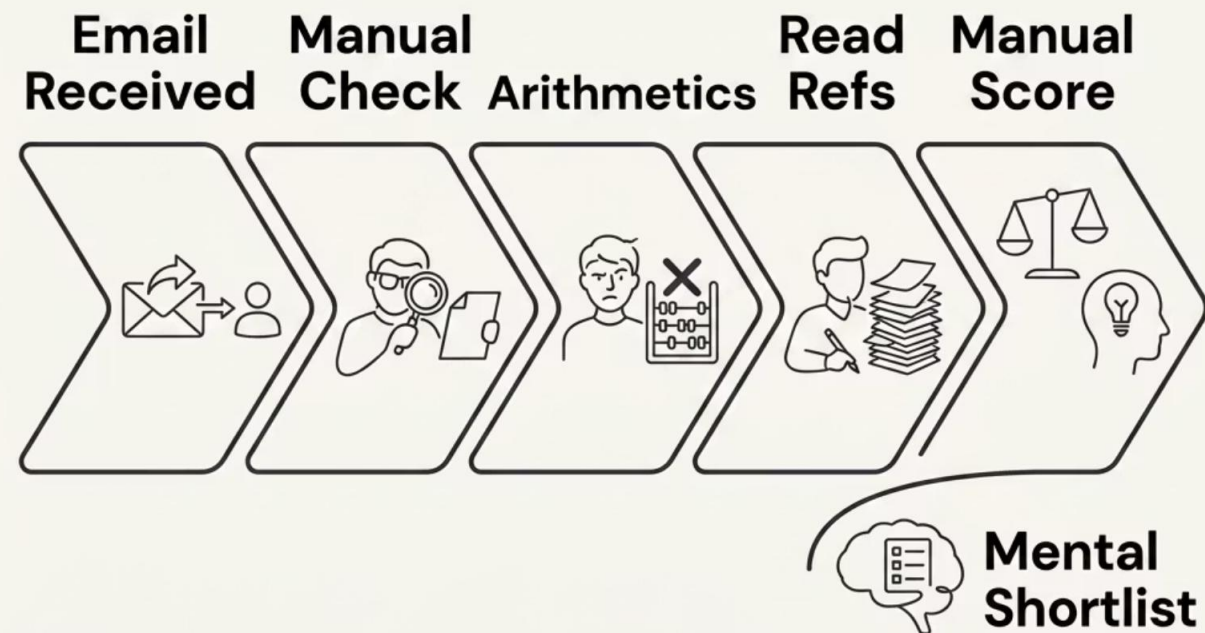
Target Review

With automated shortlist

The goal: a hybrid system where RPA handles rule-based tasks and an AI agent handles qualitative analysis, tested on 30 synthetic submissions across five Dublin properties.



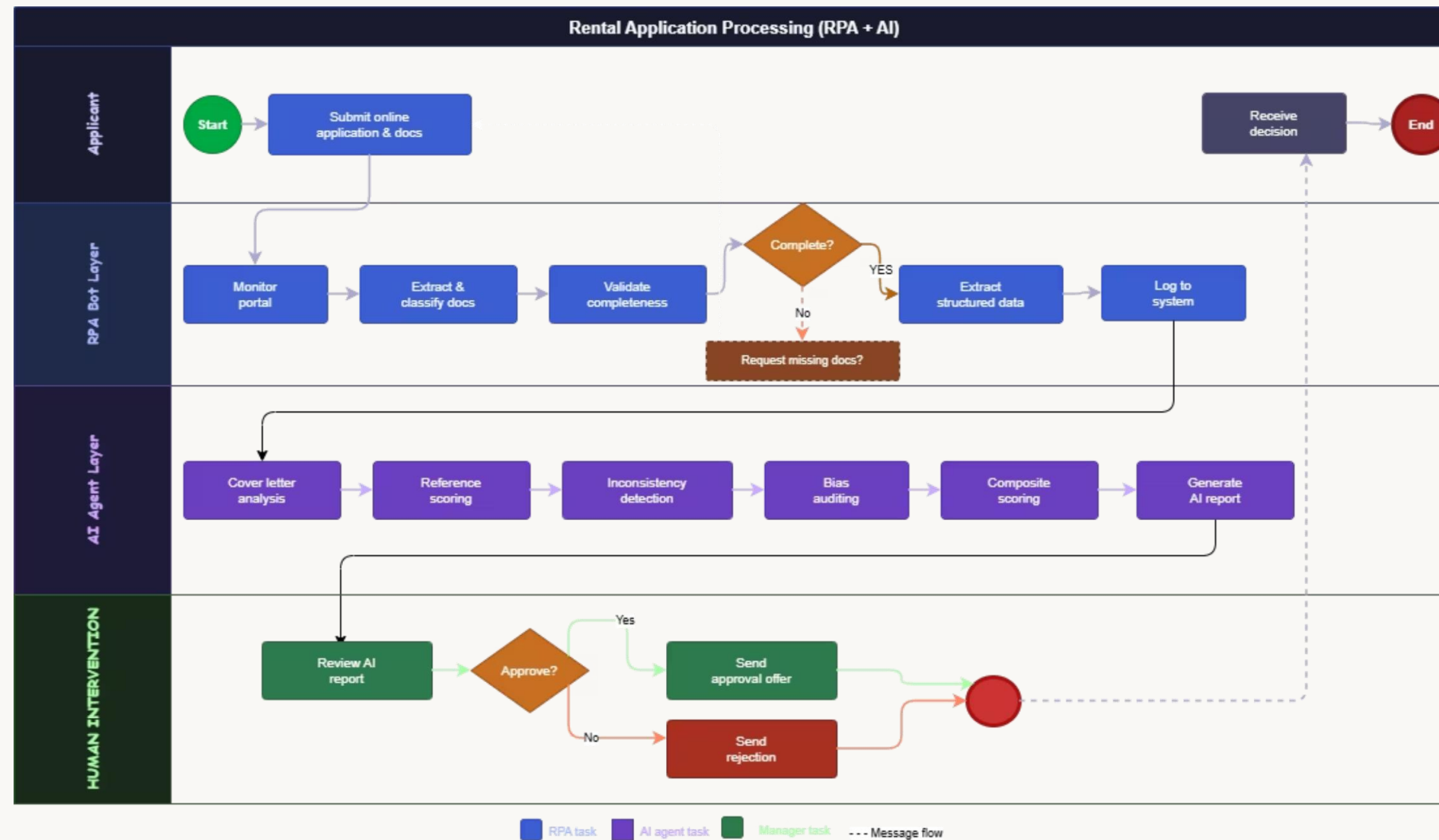
As-Is Process: Manual Workflow



Key Pain Points

- Document completeness not enforced incomplete applications slip through
- Income threshold applied inconsistently or skipped under pressure
- References read quickly, red flags missed
- No auditable record of decisions, legal exposure under Equal Status Act 2000

To-Be Process: Automated Pipeline



BPMN Diagram

The redesigned process introduces four swim lanes with clearly defined accountability: RPA enforces rules consistently, the AI layer provides qualitative intelligence, and the human agent carries legal weight of the final decision.

Automation Potential Analysis

#	Task	Layer	Justification
1–9	Intake, validation, income check, GDPR logging	RPA	Deterministic, rule-based, low risk
10–15	Reference scoring, cover letter analysis, inconsistency detection, composite score, fairness audit	AI	Requires NLP for tone, hedging, and nuance
16–18	Viewing decision, coordination, tenancy sign-off	Human	Legal liability under Equal Status Act 2000

LLMs catch signals keyword rules miss, e.g., "improved significantly" implies past problems.

RPA Layer: Rules-Based Automation

```
[DOC CHECK] REF-2026-003 [OK] All documents present
[DOC CHECK] REF-2026-004 [OK] All documents present
[DOC CHECK] REF-2026-005 [OK] All documents present
[DOC CHECK] REF-2026-006 [OK] All documents present
[DOC CHECK] REF-2026-007 [OK] All documents present
[DOC CHECK] REF-2026-008 [FAIL] Missing: landlord_reference.txt
[EMAIL SIMULATION] Sent Missing Docs email to darragh.o'brien@example.com
[DOC CHECK] REF-2026-009 [OK] All documents present
[DOC CHECK] REF-2026-010 [OK] All documents present
[DOC CHECK] REF-2026-011 [OK] All documents present
[DOC CHECK] REF-2026-012 [OK] All documents present
[DOC CHECK] REF-2026-013 [OK] All documents present
[DOC CHECK] REF-2026-014 [OK] All documents present
[DOC CHECK] REF-2026-015 [OK] All documents present
[DOC CHECK] REF-2026-016 [OK] All documents present
[DOC CHECK] REF-2026-017 [OK] All documents present
[DOC CHECK] REF-2026-018 [OK] All documents present
[DOC CHECK] REF-2026-019 [OK] All documents present
[DOC CHECK] REF-2026-020 [OK] All documents present
[DOC CHECK] REF-2026-021 [OK] All documents present
[DOC CHECK] REF-2026-022 [OK] All documents present
[DOC CHECK] REF-2026-023 [OK] All documents present
[DOC CHECK] REF-2026-024 [OK] All documents present
[DOC CHECK] REF-2026-025 [OK] All documents present
[DOC CHECK] REF-2026-026 [FAIL] Missing: photo_id_present.txt
[EMAIL SIMULATION] Sent Missing Docs email to kerry.condon@example.com
[DOC CHECK] REF-2026-027 [OK] All documents present
[DOC CHECK] REF-2026-028 [OK] All documents present
```

Pipeline Flow

- Unpack ZIP, assign REF-2026-XXX
- Validate 6 required documents
- 48-hour resubmission timer if incomplete
- Income check: $\geq 3.0x$ pass, $2.75-3.0x$ flag, $< 2.75x$ auto-reject with human review notice
- Log to SQLite GDPR ledger + Excel tracker

AI Agent Layer: Qualitative Intelligence

Analysis Chain

01

Cover Letter Analysis

Motivation, stability; no penalty for non-native English

02

Reference Scoring

Employer & landlord refs scored 1–10 with red flag detection

03

Inconsistency Detection

Cross-doc check on income, dates, reasons for moving
(Low/Medium/High)

04

Composite Score & Summary

Weighted formula + 150-word narrative per applicant

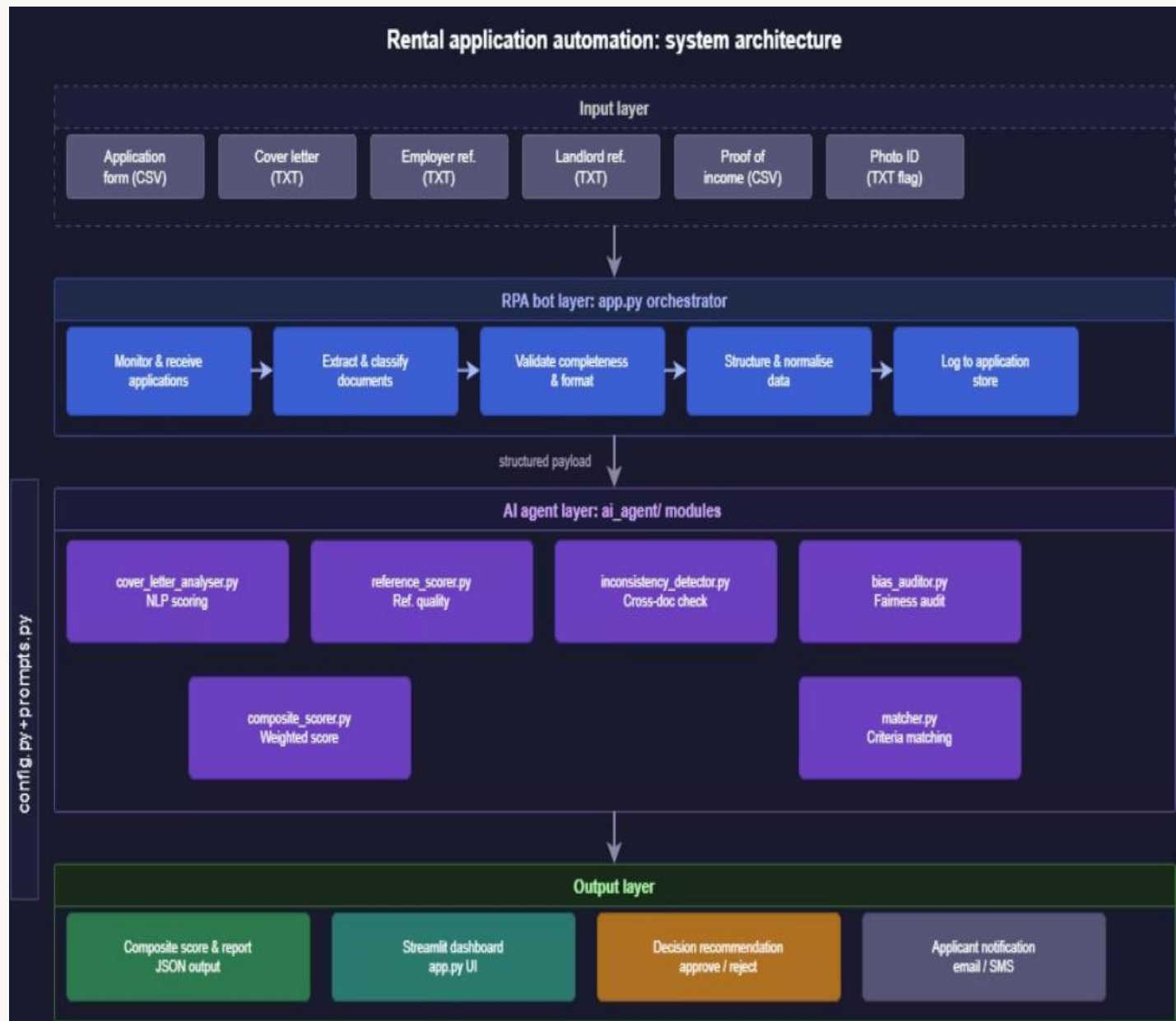
Composite Score Formula

$$S = 0.25S_{\text{inc}} + 0.20S_{\text{emp}} + 0.15S_{\text{land}} + 0.15S_{\text{cl}} + 0.15S_{\text{cons}} + 0.10S_{\text{docs}}$$

Locally hosted LLaMA 3 via Ollama; all outputs validated as structured JSON.

System Architecture & Dashboard

Six-Screen Streamlit Dashboard



System Dashboard

Portfolio summary: 5 properties, 30 applicants

Property Matching

Ranked candidates per property (top score: 87.5/100)

Applicant Deep Analysis

Full score breakdown + AI narrative

Email Draft Terminal

Context-aware drafts for 4 message types

Intake Hub

Batch ZIP upload up to 200 MB

Asset Manager

Property register with rent, beds, requirements

Ethical & Legal Considerations

GDPR Article 22

Composite scores are **recommendations only**. Final decisions stay with the human agent. All rejection emails include a right-to-review notice.

GDPR Article 13

Applicants notified of automated processing. Consent records stored in SQLite with **90-day deletion** schedule.

Equal Status Act 2000

Bias audit checks for proxy discrimination via writing style or language complexity. Cover letter prompts explicitly avoid penalising non-native English.

EU AI Act, High Risk

Housing access AI is high-risk under Annex III. Post-hoc audit, human-in-the-loop gate, and transparency notices form the minimum compliance architecture.



Conclusion & Limitations

What Was Achieved

- Hybrid RPA + AI system fully implemented in Python
- Tested on 30 synthetic submissions across 5 Dublin properties
- Agent workload reduced from **6 to 8 hours to ~20 minutes**
- GDPR logging, transparency disclosures, and bias audit built in from the start
- Every score fully traceable via structured JSON output

Limitations & Future Work

- Strict document schema requires a structured web intake form in production
- LLM output variability mitigated but not eliminated by JSON validation
- Future: NLP-based classifier for unstructured email submissions
- Future: Fine-tuned model trained on Irish rental application language